Part 9
Cenozoic Diversification of Animals

9.5 Primates

EART11100: How to Grow a Planet
Robert Sansom
University of Manchester



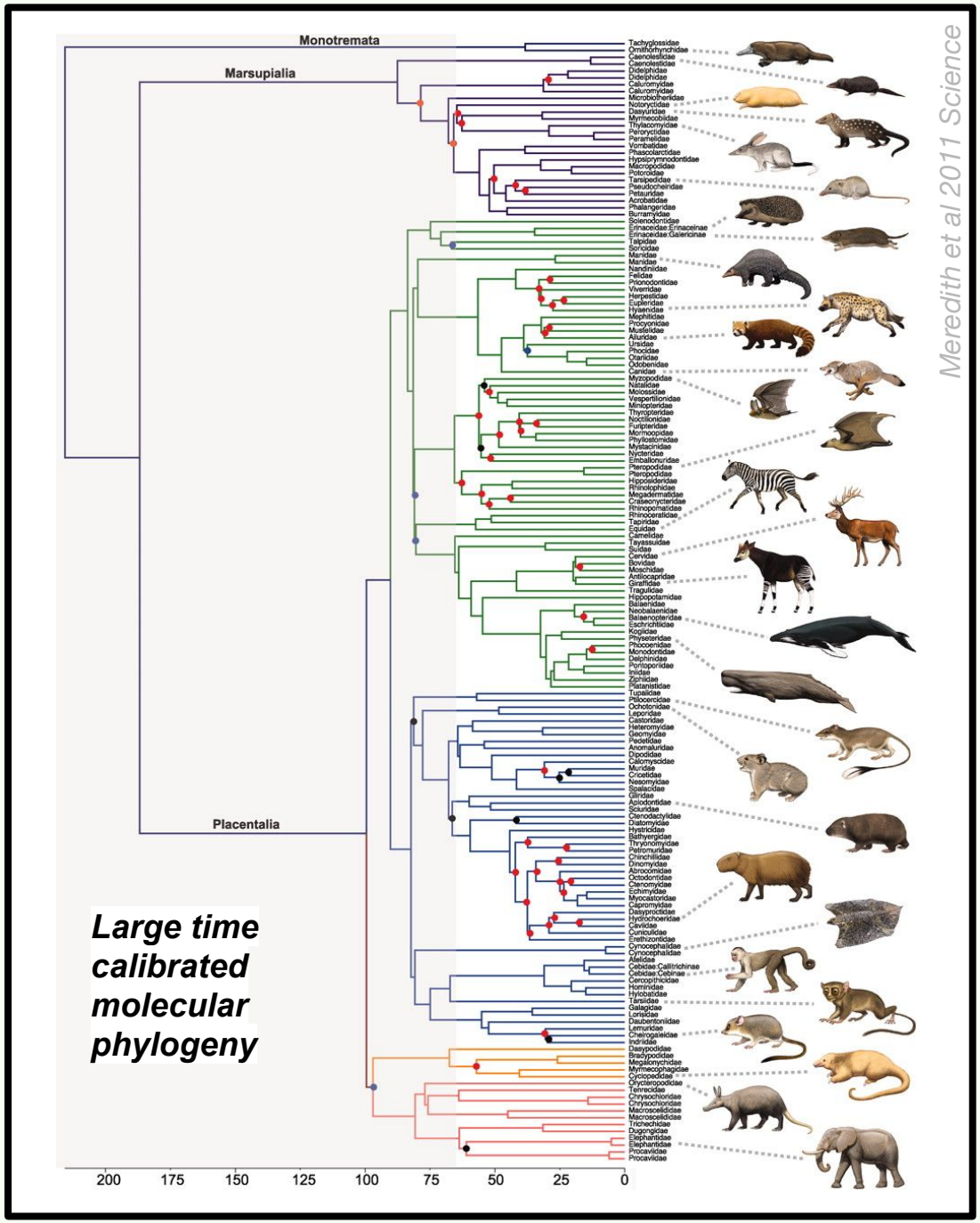
Didier Descouens

Palaeogene Summary

- Diversification of vertebrates after the end Cretaceous extinction
- Warm conditions in the Eocene, particularly the thermal maximum
- Spread of grasses



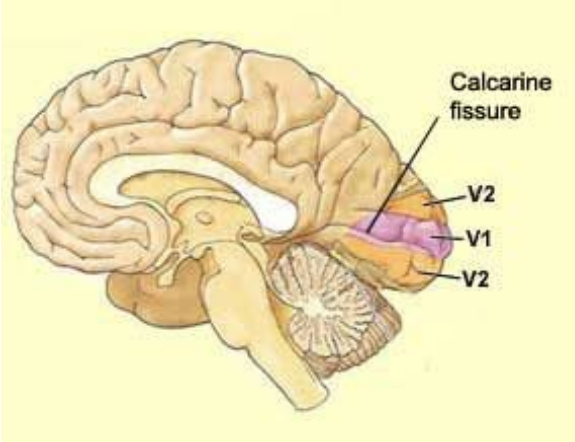
Mammal Evolution



Primates



Primates are a diverse group of mammals with relatively large brains, specialized for vision, and tactile grasping feet and hands





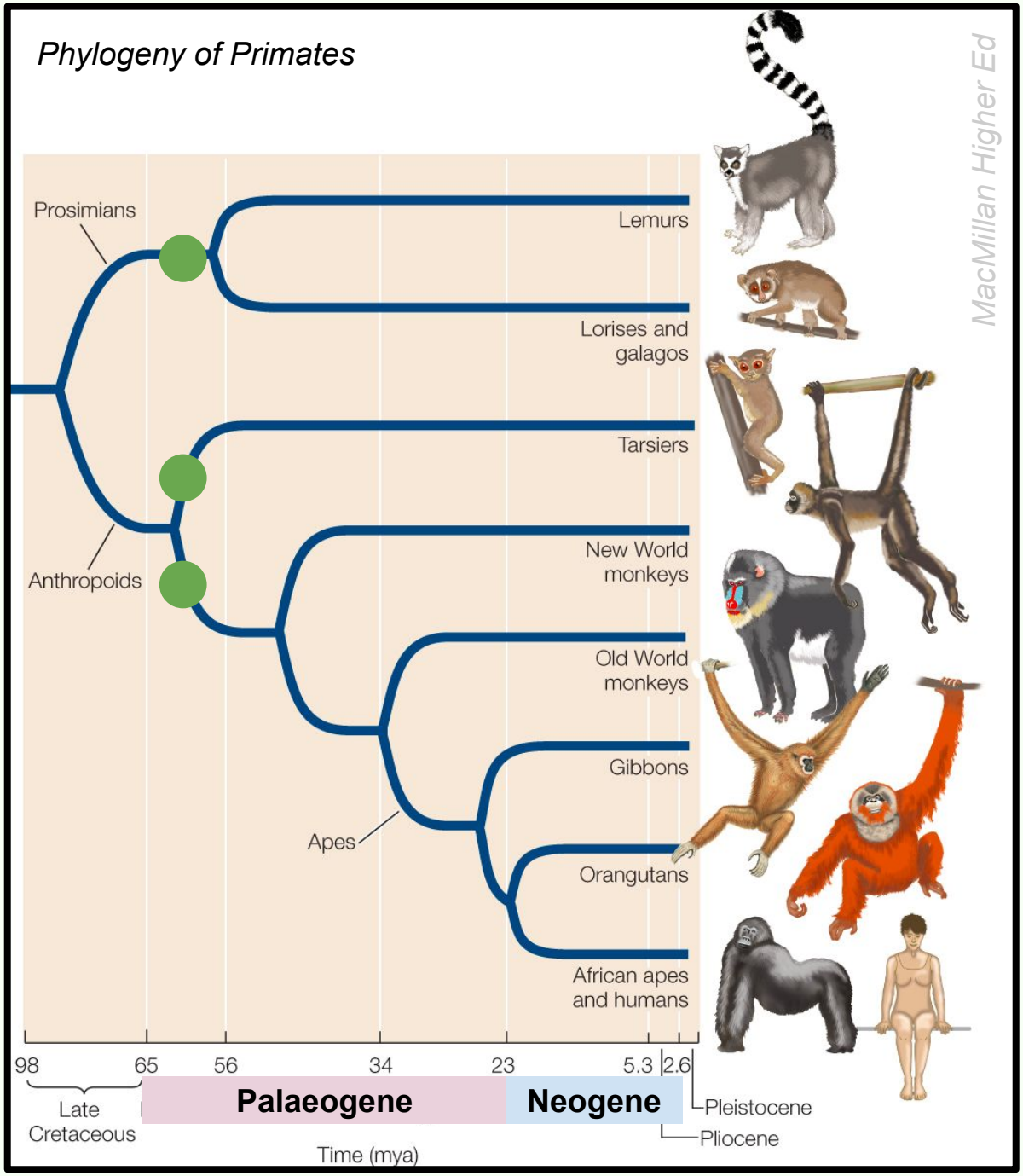
First diversified in Palaeogene, including the lemur-like *Darwinus*

Primates

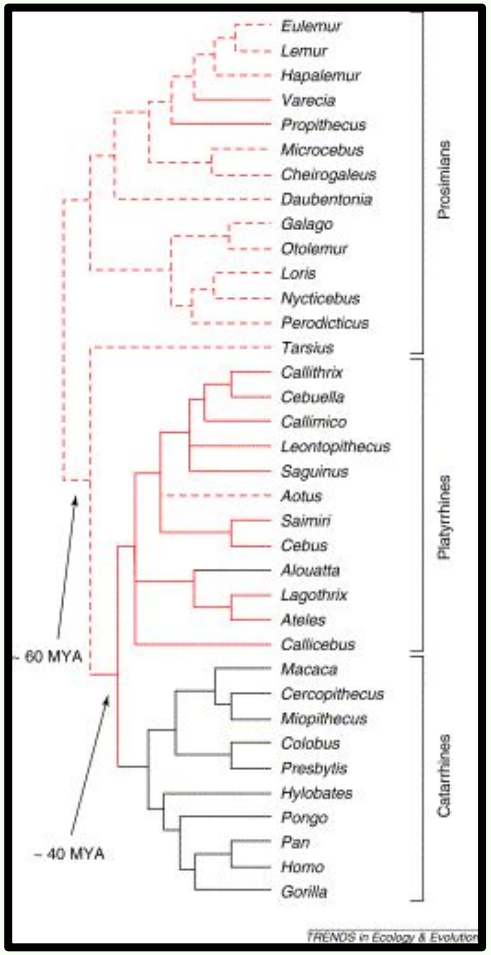
Primates have a long evolutionary history, divergences taking place in Palaeogene

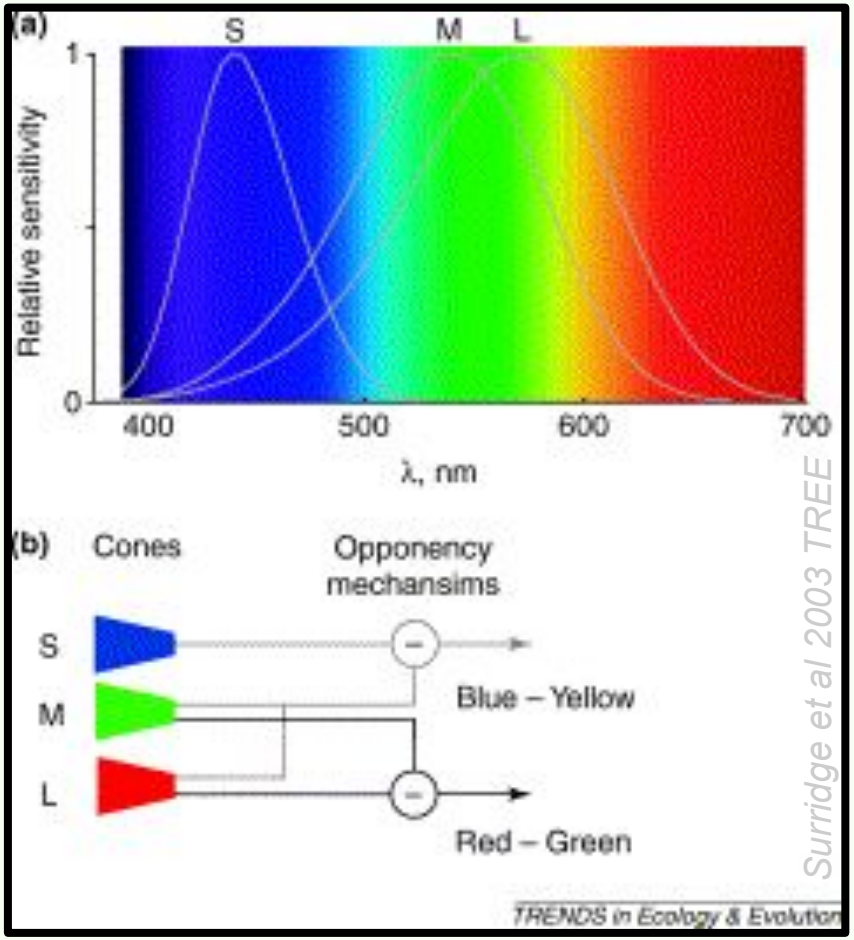
Group comprises of Lemuriformes, Tarsiiformes, and Simiiformes

Primary specializations of primates relate to vision and thumbs



Colour Vision

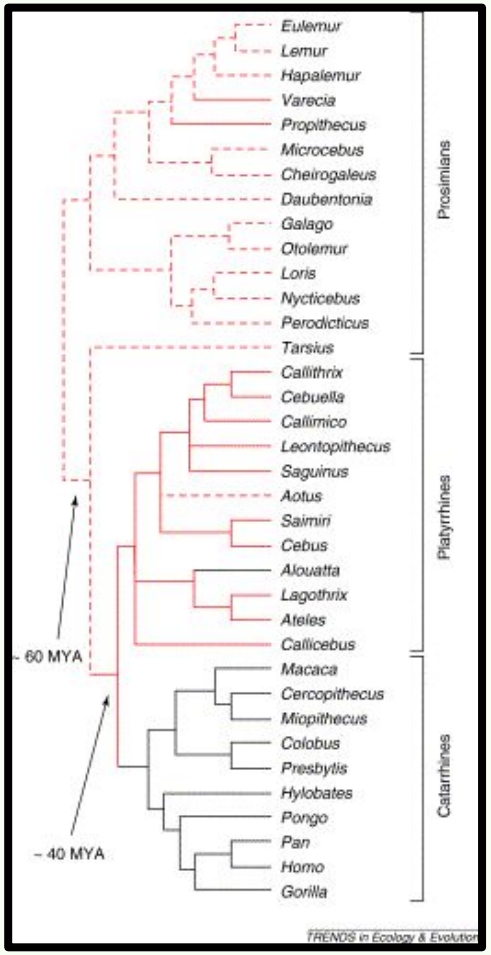


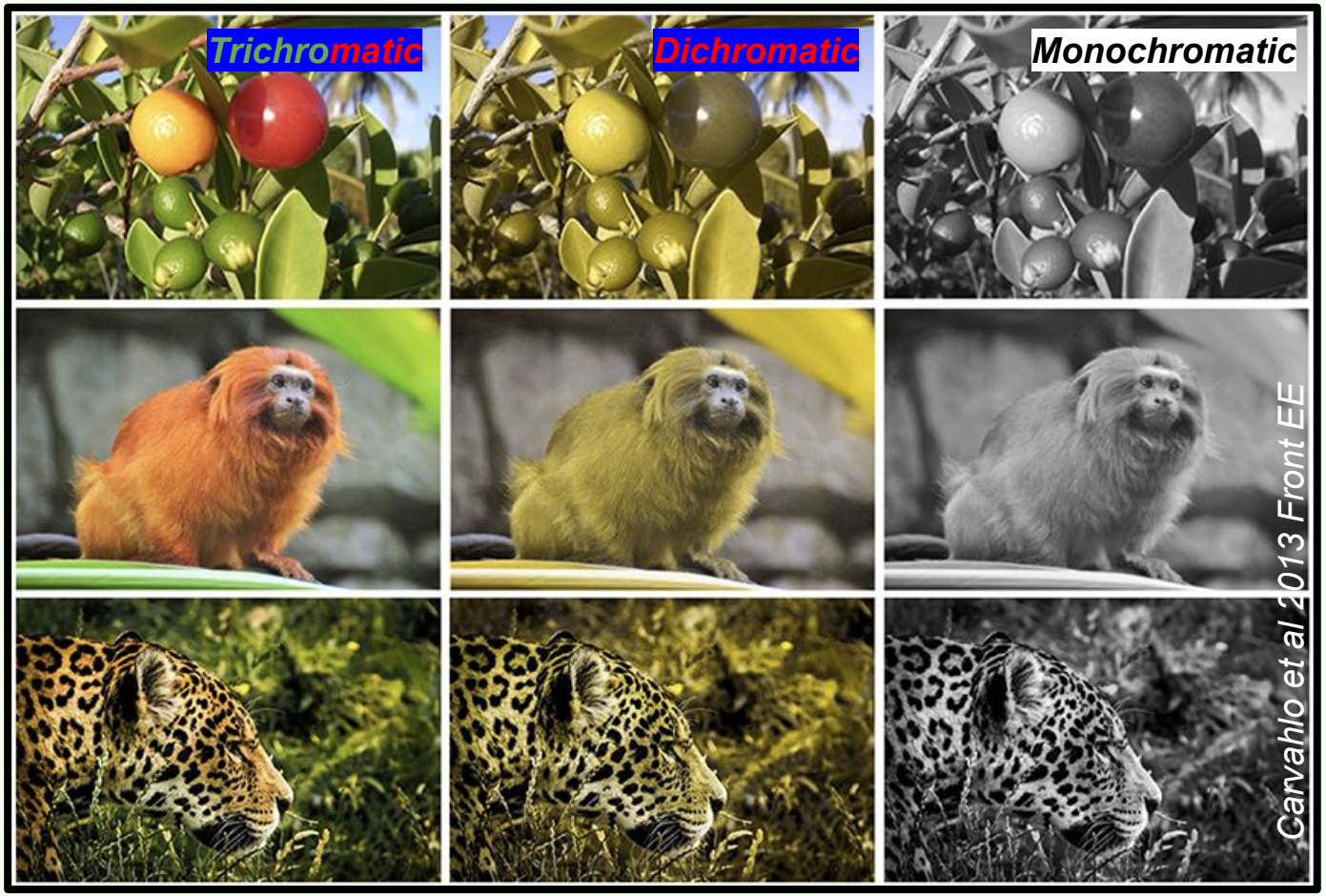


Most mammals have either monochromatic or dichromatic vision

Old world monkeys and apes have trichromacy (red, green and blue cone cells)

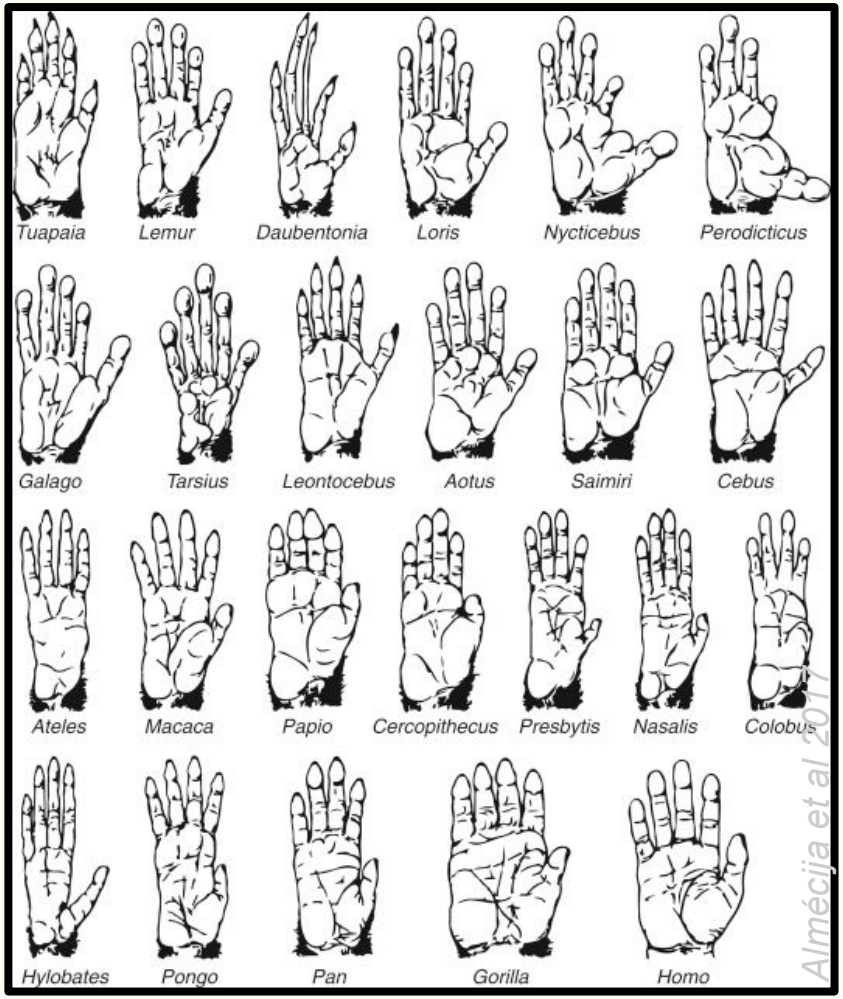
Colour Vision

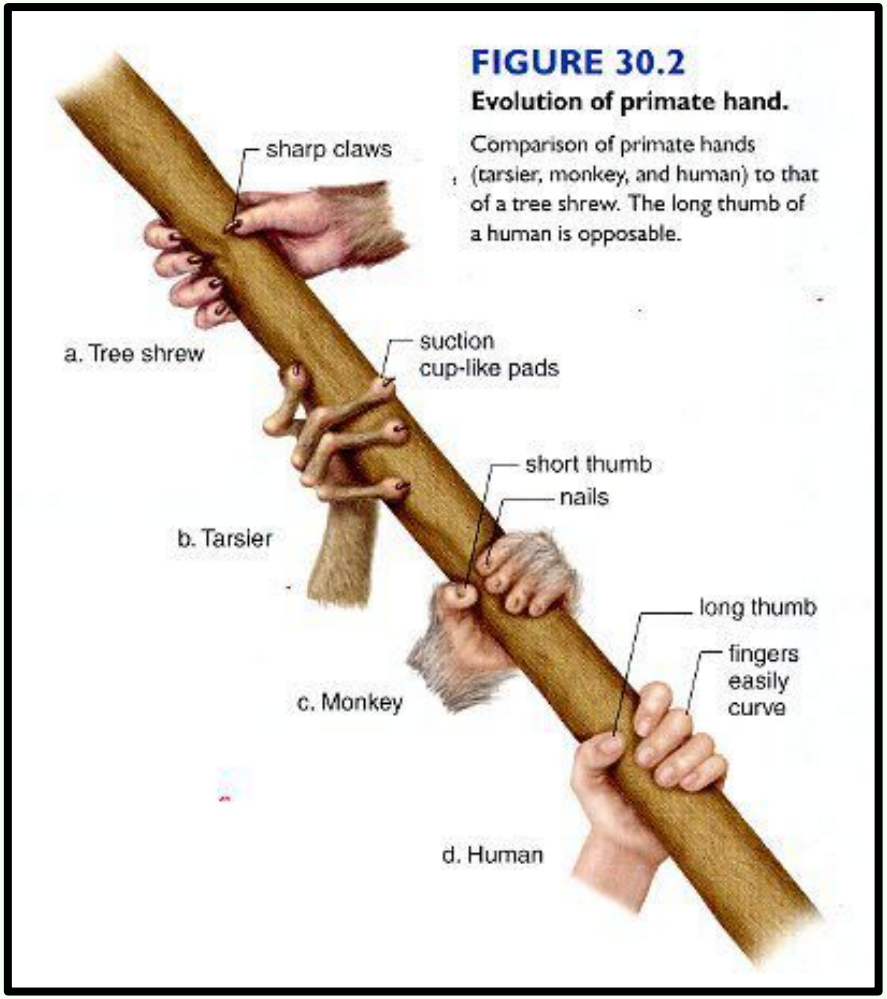




Grasping Hands

Primate hands show increasingly complexity of grasping due to flat keratin nails, opposable thumbs

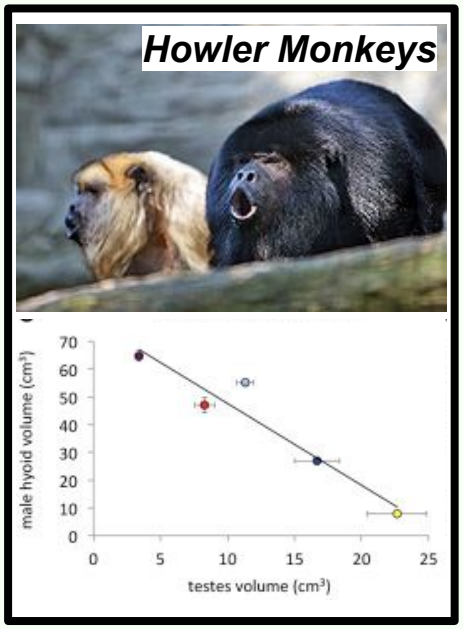




Sexual Dimorphism

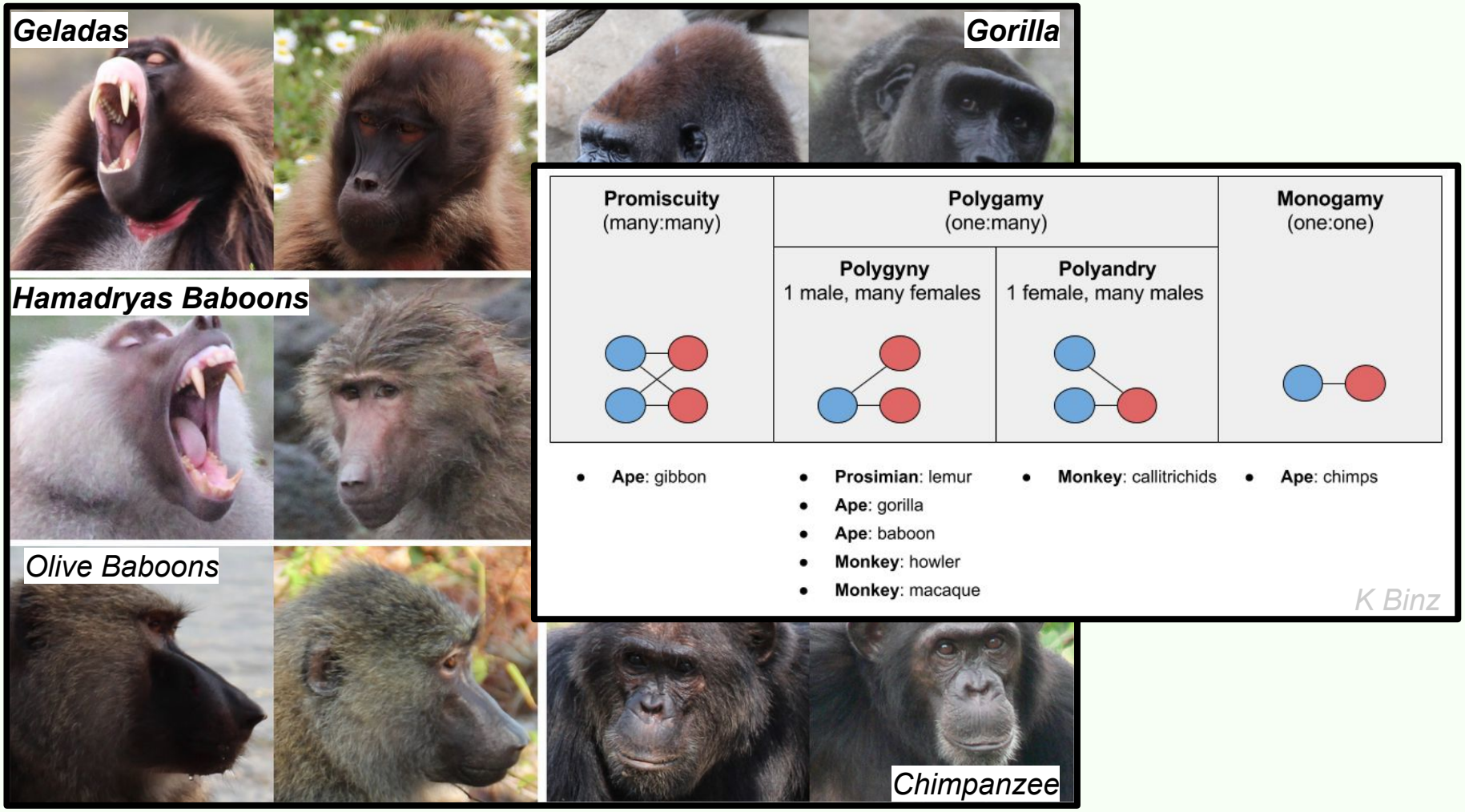


Wilson et al 2017 PRSB



Dunn et al 2015
Current Biology

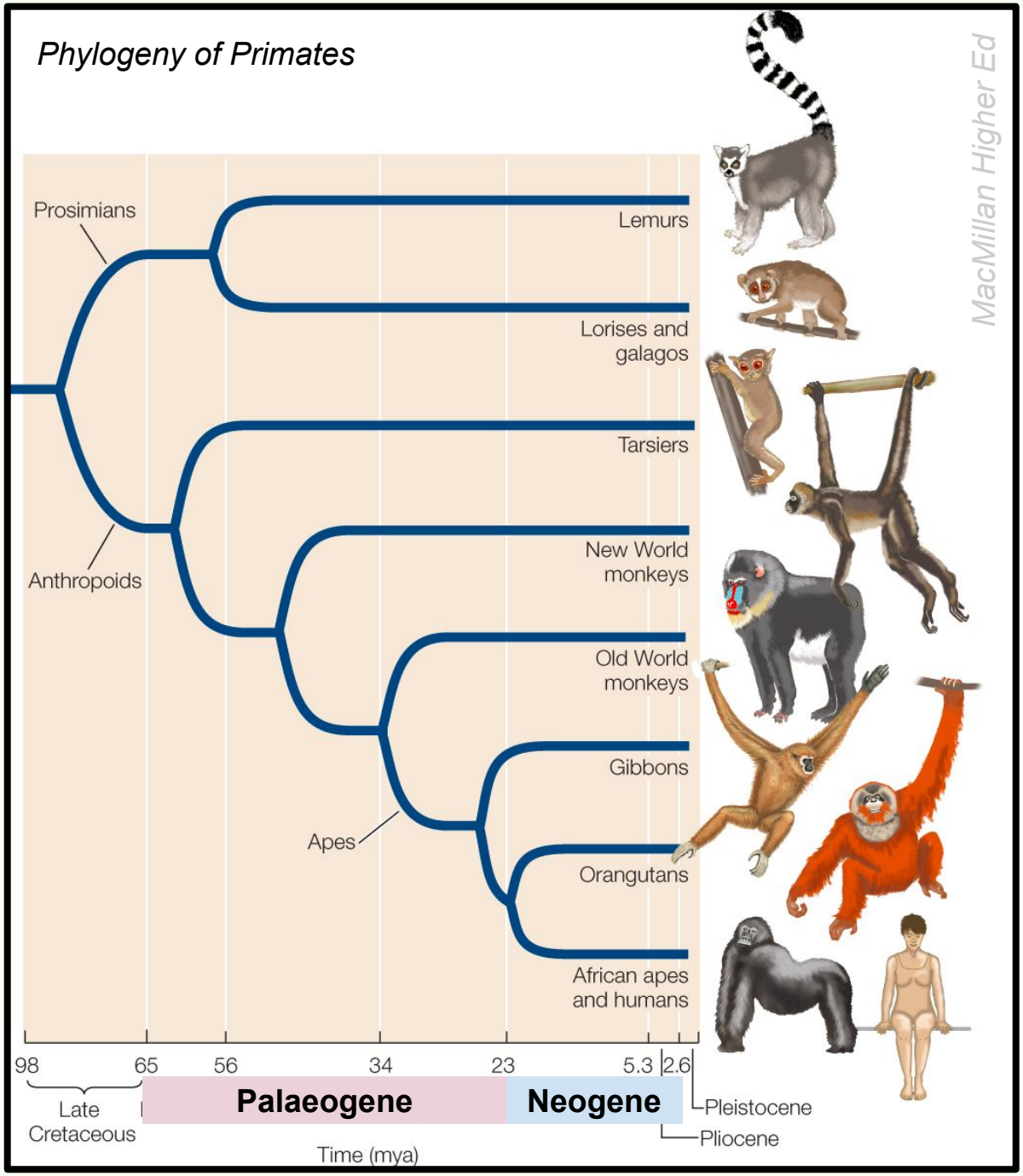
Sexual Dimorphism



Wilson et al 2017 PRSB

Primates

Within simiiformes, there are new world monkeys, old world monkeys and apes

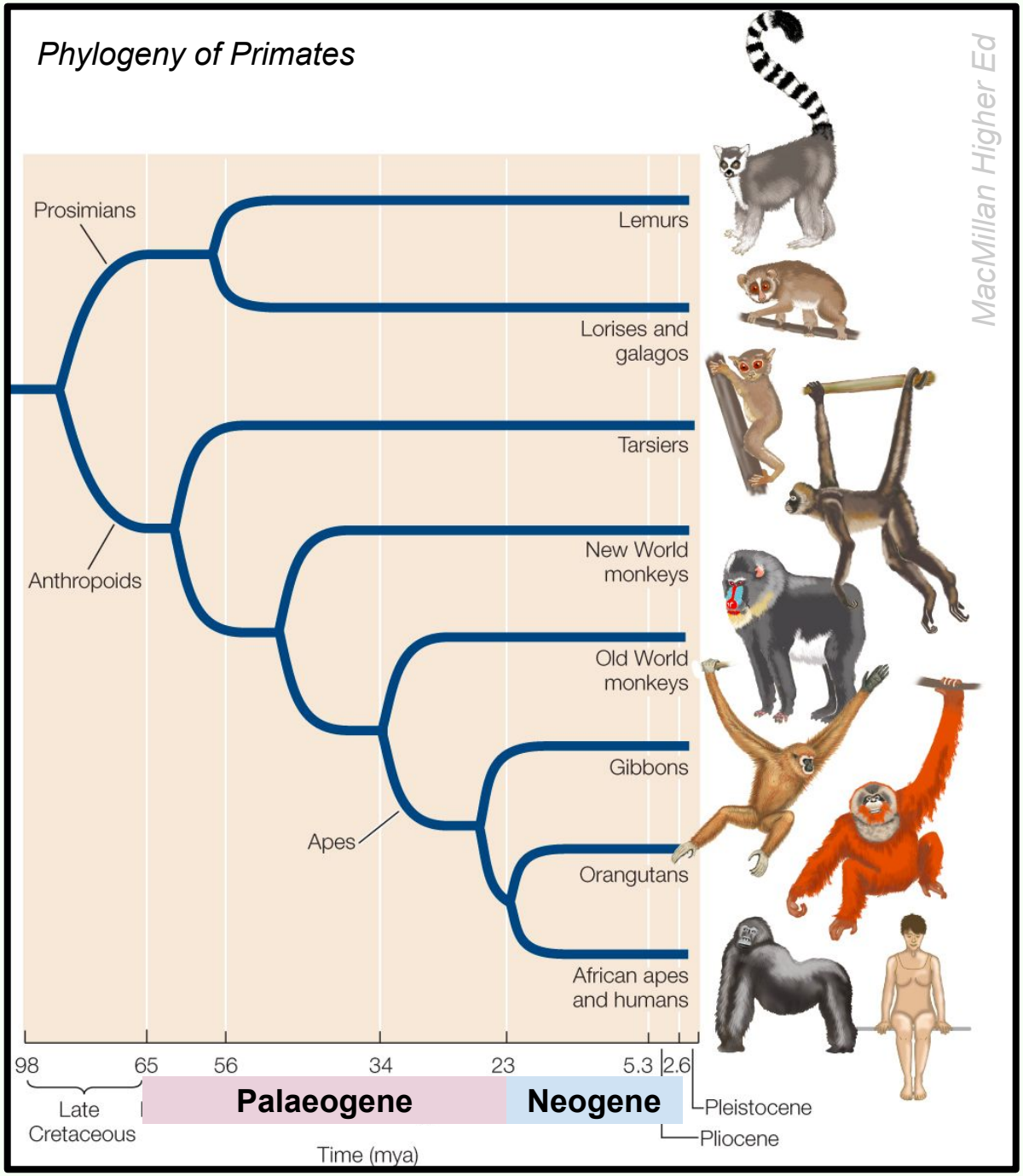


Primates

Within primates, we are part of the **Hominidae** (Great Apes), primates which have lost their tails, and exhibit complex behaviours



Living non-human species of Hominidae



Great Apes

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Living non-human species of Hominidae



Chimpanzee



Monkey

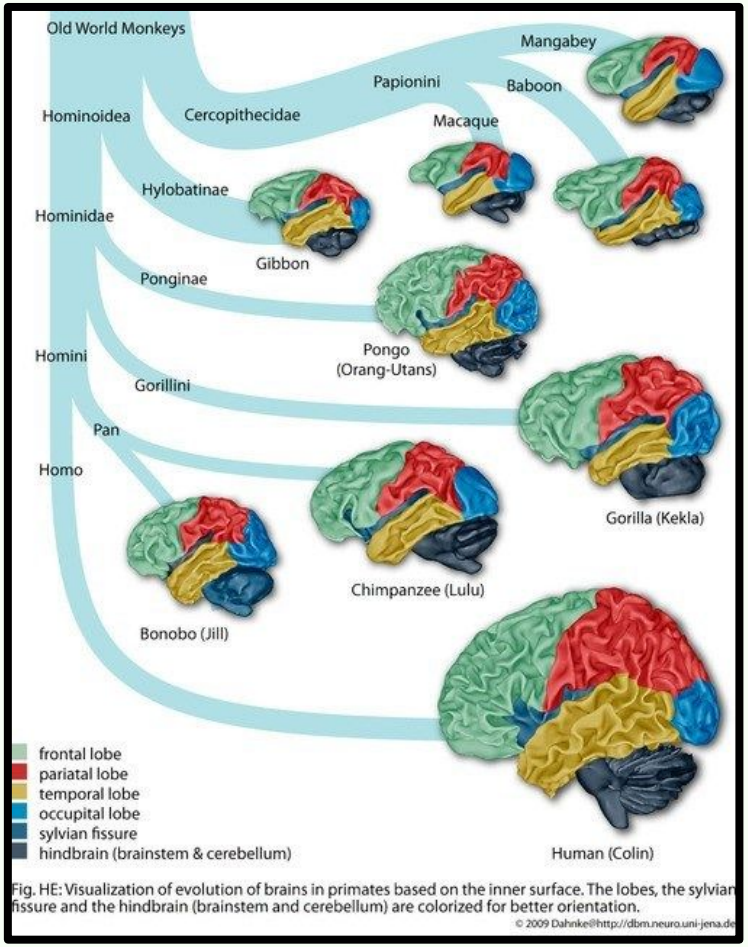
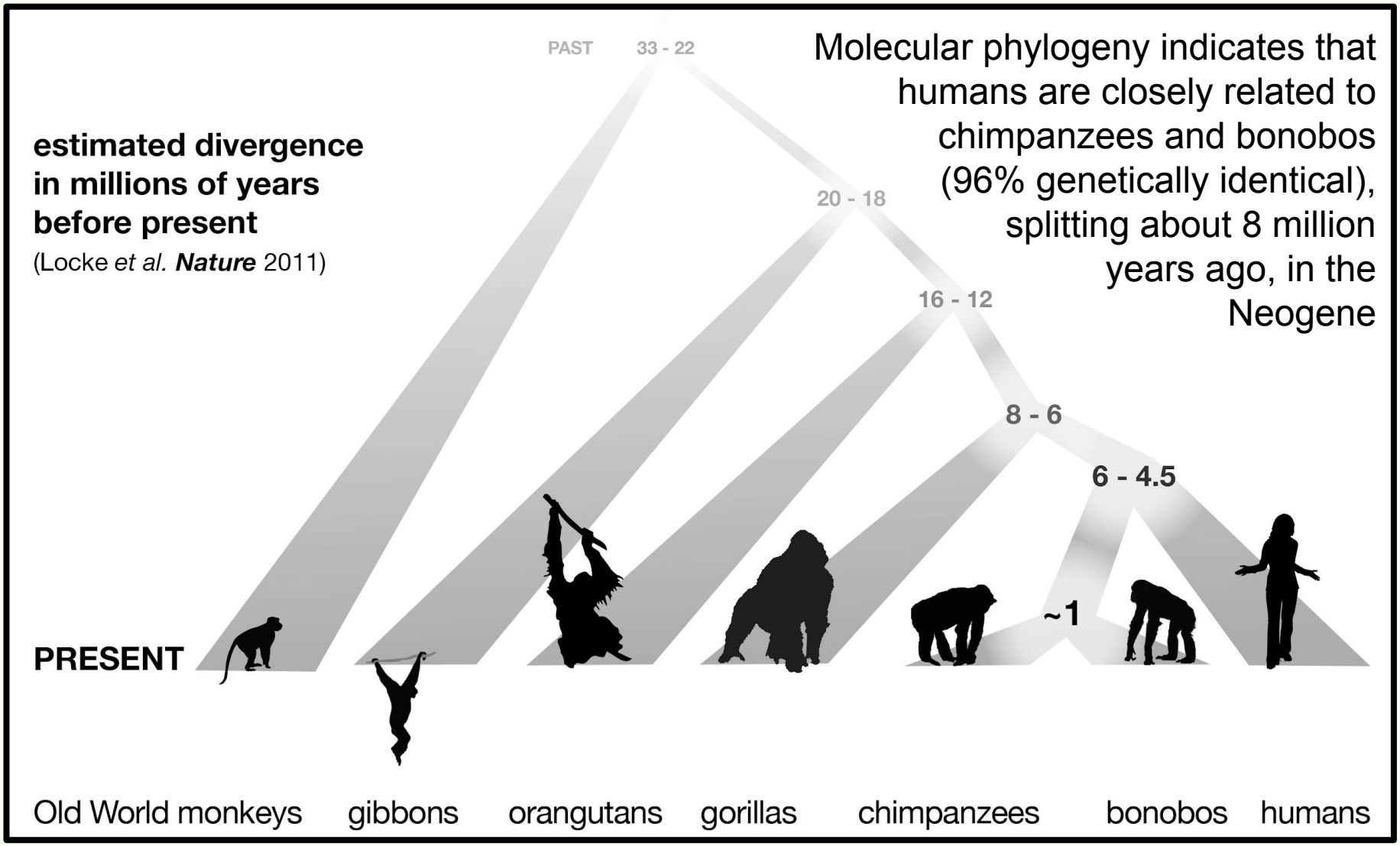


Fig. HE: Visualization of evolution of brains in primates based on the inner surface. The lobes, the sylvian fissure and the hindbrain (brainstem and cerebellum) are colorized for better orientation.

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Primate Evolution



Cenozoic Diversification of Animals

9.5 Primates

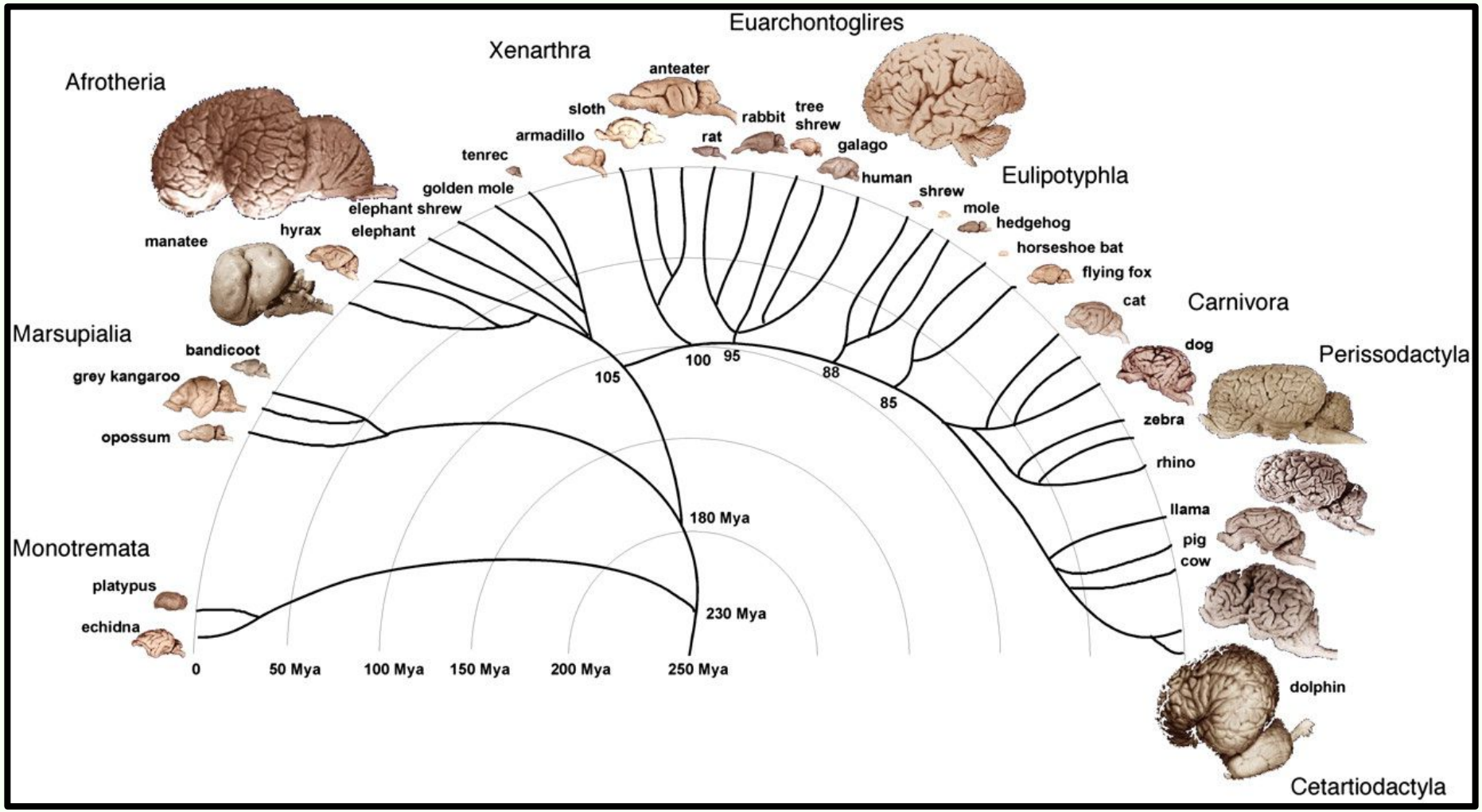
Intended Learning Objective: Outline the biology of primates in the context of human evolution

Associated Task: Take a look at the phylogeny of mammal brains and their sizes below the video. What factors might be related to the evolution of big brains?

Robert Sansom, U. Manchester



Didier Descouens


Euarchontoglires
Xenarthra
Afrotheria
anteater
sloth
armadillo
tenrec
rabbit
rat
tree shrew
galago
human
Eulipotyphla
shrew
mole
hedgehog
horseshoe bat
flying fox
cat
Carnivora
dog
zebra
Perissodactyla
rhino
llama
pig
cow
dolphin
Cetartiodactyla
golden mole
elephant shrew
elephant
hyrax
manatee
Marsupialia
bandicoot
grey kangaroo
opossum
Monotremata
platypus
echidna
105
100
95
88
85
180 Mya
230 Mya
0
50 Mya
100 Mya
150 Mya
200 Mya
250 Mya